

AC3303

Introduction to Asset Pricing & Valuation

Spring 2024 Examination

Paper 1 · 1.5 hours · Answer Question 1 and one other question · Total 100 Marks

Module	AC3303 — Introduction to Asset Pricing & Valuation
Session	Spring 2024
Paper	1 of 1
External Examiner	Professor Thomas O'Connor
Internal Examiner	Dr. Joe Healy
Structure	Question 1 (Compulsory, 50 Marks) + one of Question 2 or Question 3 (50 Marks)
This scheme covers	Question 1, Question 2 and Question 3 in full, so you can study either route through the paper

Own-figure rule. Where a candidate makes an error early in a multi-step question, full method marks are still awarded for correctly applying the required technique to their own, incorrect figure in every subsequent step. This applies throughout every question in this scheme and is restated wherever it is especially relevant.

How this scheme was built. Every calculation below was computed independently in Python before being written up and cross-checked against internal consistency tests. Discursive answers draw on standard corporate finance theory rather than any one source, with own-figure and alternative-argument credit noted throughout.

Question 1 (Compulsory)

50 Marks — Portfolio Theory & Efficient Market Hypothesis

Scenario

CE Bank Group Plc and FK Capital Group Plc are two Financial Services companies trading on the Euronext in Dublin. Their quarterly returns for 2023:

Quarter	CE Bank Group	FK Capital Group
Q1 2023	12%	4%
Q2 2023	-3%	4%
Q3 2023	7%	-1%
Q4 2023	20%	5%

(a) Required

6 Marks

Calculate both the expected return and standard deviation risk for the two companies.

FULL MODEL ANSWER

This is a historical sample of four quarterly returns rather than a probability-weighted scenario table. Each of the four quarters is treated as an equally likely outcome (probability 0.25 each), which mirrors how expected return and risk are computed from a probability distribution elsewhere in this course — the expected return is the simple average of the four quarterly returns, and the variance is the average squared deviation from that mean.

Quarter	Return CE	Dev. from mean	Squared × 0.25	Return FK	Dev. from mean	Squared × 0.25
Q1 2023	12%	3.00	2.25	4%	1.00	0.25
Q2 2023	-3%	-12.00	36.00	4%	1.00	0.25
Q3 2023	7%	-2.00	1.00	-1%	-4.00	4.00
Q4 2023	20%	11.00	30.25	5%	2.00	1.00
Mean / Variance	9.00%		69.50	3.00%		5.50

$E(R_{CE}) = 9.00\%$, $SD(CE) = 8.34\%$ $E(R_{FK}) = 3.00\%$, $SD(FK) = 2.35\%$

Rounding/methodology note. A candidate who instead used the sample (n–1) standard deviation convention — standard in general statistics for a small historical sample — would get $SD(CE) = 9.63\%$ and $SD(FK) = 2.71\%$. Either convention is acceptable and should be credited (own-figure carried through to parts b–d), provided it is applied consistently across both companies and into every subsequent part.

Mark Allocation — Part (a), 6 Marks

Tier	Marks	What separates this tier
6	6	Both expected returns and both standard deviations correct with clear working, using a consistent variance convention.
4-5	4-5	Both expected returns correct; one standard deviation correct, the other has a minor slip (own-figure).
2-3	2-3	Correct method but a computational error affecting more than one figure.
0-1	0-1	Wrong technique or no workable attempt.

Question 1 (Compulsory)

50 Marks — Portfolio Theory & Efficient Market Hypothesis

(b) Required

6 Marks

Calculate the covariance between returns on CE Bank Group Plc and FK Capital Group Plc.

FULL MODEL ANSWER

Quarter	Dev. CE	Dev. FK	Product	× 0.25
Q1 2023	3.00	1.00	3.00	0.75
Q2 2023	-12.00	1.00	-12.00	-3.00
Q3 2023	-2.00	-4.00	8.00	2.00
Q4 2023	11.00	2.00	22.00	5.50
Covariance				5.25

$Cov(CE, FK) = 5.25$

The positive but modest covariance suggests the two companies' returns move together to some extent — plausible for two firms in the same sector (Financial Services) — but far less closely than the near-perfectly correlated pairs seen in other papers in this series.

Mark Allocation — Part (b), 6 Marks

Tier	Marks	What separates this tier
6	6	Correct covariance with full working shown.
4-5	4-5	Correct method but a minor own-figure carry-through error from part (a).
1-3	1-3	Some correct elements but incorrect combination or formula.
0	0	No credible attempt.

Question 1 (Compulsory)

50 Marks — Portfolio Theory & Efficient Market Hypothesis

(c) Required

7 Marks

Calculate the correlation coefficient between the returns on CE Bank Group Plc and FK Capital Group Plc. What are the implications of this level of correlation for portfolio risk?

FULL MODEL ANSWER

$$\rho = \text{Cov} \div (\text{SD}_{\text{CE}} \times \text{SD}_{\text{FK}}) = 5.25 \div (8.34 \times 2.35) = 0.269$$

Strength and direction: A correlation of 0.269 is positive but weak-to-moderate — the two shares' returns move in the same general direction only loosely, in clear contrast to the near-perfect correlations (0.9+) seen in some other scenario-based questions in this series.

Implication for portfolio risk: Because correlation is well below +1, combining CE Bank Group and FK Capital Group in a portfolio provides a genuine diversification benefit — the portfolio standard deviation will fall meaningfully below the simple weighted average of the two individual standard deviations, since the low correlation means the two shares' return fluctuations partly offset one another rather than reinforcing each other.

Practical takeaway: This is a much more useful pairing for risk reduction than two highly-correlated shares, since an investor holding both can reduce total portfolio risk without necessarily sacrificing expected return, up to the point of the minimum-variance combination of the two.

Mark Allocation — Part (c), 7 Marks

Tier	Marks	What separates this tier
6-7	6-7	Correct correlation coefficient (own-figure) plus a clear explanation of the diversification implication, correctly noting the benefit is stronger here than for highly-correlated pairs.
4-5	4-5	Correct coefficient with a partial explanation of the diversification implication.
2-3	2-3	Correct method but computational error, or correct coefficient with minimal explanation.
0-1	0-1	Incorrect formula or no attempt.

Question 1 (Compulsory)

50 Marks — Portfolio Theory & Efficient Market Hypothesis

(d) Required7 Marks

Determine the portfolio expected return and standard deviation risk if 55% of the fund is invested in CE Bank Group Plc and the remainder is invested in FK Capital Group Plc. Is this portfolio efficient?

FULL MODEL ANSWER

Component	Value
Weight in CE Bank Group (w_1)	0.55
Weight in FK Capital Group (w_2)	0.45
$E(R_p) = 0.55(9.00\%) + 0.45(3.00\%)$	6.30%
$w_1^2\sigma_1^2 = 0.55^2 \times 69.50$	21.0238
$w_2^2\sigma_2^2 = 0.45^2 \times 5.50$	1.1138
$2w_1w_2Cov = 2 \times 0.55 \times 0.45 \times 5.25$	2.5988
Portfolio Variance σ_p^2	24.7363
Portfolio Standard Deviation σ_p	4.97%

Diversification confirmed. The simple weighted average of the two individual standard deviations would be $0.55(8.34\%) + 0.45(2.35\%) = 5.64\%$ — the actual portfolio SD of 4.97% sits below this, confirming a genuine diversification benefit from the correlation being well under +1 (as flagged in part (c)).

Question 1 (continued)

Is this portfolio efficient?

Finding the minimum-variance combination: For a two-asset portfolio, the minimum-variance weight in CE Bank Group is $w^* = (\sigma_{FK}^2 - \text{Cov}) \div (\sigma_{CE}^2 + \sigma_{FK}^2 - 2\text{Cov}) = (5.50 - 5.25) \div (69.50 + 5.50 - 2(5.25)) = 0.25 \div 64.50 \approx \mathbf{0.4\% \text{ in CE Bank Group}}$ (essentially all-FK), giving a minimum portfolio SD of about 2.35%.

Where the 55/45 portfolio sits on the frontier: The two-asset efficient frontier runs from this minimum-variance point ($\approx 100\%$ FK) up through increasing weight in the higher-return, higher-risk asset (CE Bank Group) to 100% CE Bank Group. Since the 55/45 portfolio holds far more than the 0.4% minimum-variance weight in CE Bank Group, it sits on the **upper, efficient branch** of this two-asset frontier — for these two assets in isolation, there is no alternative combination that offers a higher expected return for the same or lower risk.

Important scope caveat. This conclusion holds only within the two-asset universe of CE Bank Group and FK Capital Group. True portfolio efficiency is defined relative to the full investable universe (all available risky assets, plus a risk-free asset) — a candidate who notes that this two-asset comparison cannot confirm efficiency in the broader Markowitz/CAPM sense should receive full credit for that observation, alongside the two-asset frontier argument above.

Mark Allocation — Part (d), 7 Marks

Tier	Marks	What separates this tier
6-7	6-7	Correct portfolio return and SD using the full variance-covariance formula, AND a reasoned discussion of efficiency (ideally referencing the minimum-variance point or the scope caveat).
4-5	4-5	Correct portfolio return and SD but efficiency discussion superficial (e.g. simple 'yes' or 'no' with no reasoning).
2-3	2-3	Portfolio return correct only, or variance calculated without the covariance term; efficiency not addressed.
0-1	0-1	No credible attempt.

Question 1 (Compulsory)

50 Marks — Portfolio Theory & Efficient Market Hypothesis

(e) Required**24 Marks**

In the context of the Efficient Markets Hypothesis, explain briefly what is meant by 'Stock Market Efficiency'. In addition, describe the 'semi-strong' level of stock market efficiency in detail. Discuss the research that either supports or challenges this level of stock market efficiency.

FULL MODEL ANSWER

Stock market efficiency (the Efficient Market Hypothesis) holds that security prices fully and rapidly reflect all available information, such that no investor can consistently earn abnormal (risk-adjusted) returns by trading on that information. Three forms are conventionally distinguished by the information set assumed to be reflected in price: weak form (past price/volume data), semi-strong form (all publicly available information), and strong form (all information, including private/insider information).

The semi-strong form in detail

Semi-strong form efficiency states that share prices reflect not only past price data but **all publicly available information** — published accounts, earnings announcements, dividend declarations, macroeconomic data, analyst forecasts, and any other information released to the public. Under this form, fundamental analysis of publicly available information cannot generate abnormal returns, because by the time an investor has read and analysed public information, the market has already incorporated it into the price. The standard test of semi-strong efficiency is the **event study**: measuring how quickly and accurately prices adjust around the public release of new information (e.g. an earnings announcement), and checking whether any abnormal return persists after the announcement (which would indicate inefficiency) or is fully absorbed immediately (consistent with efficiency).

Question 1 (continued)

Research supporting semi-strong efficiency

Classic event studies: Fama, Fisher, Jensen and Roll (1969) studied price reactions around stock split announcements and found that abnormal returns were fully incorporated into price by the time of the split, with no exploitable abnormal return remaining afterward — widely cited as early strong evidence for semi-strong efficiency.

Rapid price adjustment more generally: A large subsequent literature of event studies around earnings announcements, dividend changes, and merger announcements generally finds that the bulk of the price reaction occurs within a very short window around the announcement, consistent with rapid incorporation of public information.

Research challenging semi-strong efficiency (documented anomalies)

Post-earnings-announcement drift (PEAD): Ball and Brown (1968) and a large subsequent literature find that stock prices continue to drift in the direction of an earnings surprise for weeks or months after the announcement, rather than adjusting instantly — a well-documented anomaly inconsistent with semi-strong efficiency.

Size and value effects: Banz (1981) documented a small-firm effect (small-cap stocks earning higher risk-adjusted returns than predicted), and Fama and French (1992) documented a book-to-market/value effect — both suggesting publicly available information (firm size, book-to-market ratio) can be used to predict returns, which should not be possible under semi-strong efficiency.

Momentum effect: Jegadeesh and Titman (1993) found that stocks with strong recent past performance continue to outperform over the following 3–12 months, an anomaly that is also difficult to reconcile with rapid, unbiased incorporation of public information.

Balanced conclusion: The overall empirical picture is mixed: markets adjust very quickly to most public information most of the time (supporting semi-strong efficiency as a reasonable working approximation), but persistent, replicated anomalies such as PEAD and momentum indicate that semi-strong efficiency does not hold perfectly or universally — an active area of ongoing research and debate.

Candidates may cite different specific studies or examples; what matters is correctly identifying at least one supporting strand of evidence and at least one challenging anomaly, each properly linked back to the definition of semi-strong efficiency, rather than reproducing this exact set of citations.

Mark Allocation — Part (e), 24 Marks

Tier	Marks	What separates this tier
19-24	19-24	Clear, accurate definition of EMH; thorough, detailed description of semi-strong form; balanced discussion of BOTH supporting and challenging research with specific, correctly-described evidence.
13-18	13-18	Good definition and semi-strong description; research discussion present but one-sided (only supporting OR only challenging evidence) or lacking specific studies.
7-12	7-12	Definition present but semi-strong form not developed in sufficient detail; little to no research evidence discussed.
0-6	0-6	Definition of EMH absent or materially incorrect; no meaningful discussion of semi-strong form or evidence.

Question 1 Total: 50 Marks

Question 2

50 Marks — Choice question: answer EITHER Question 2 OR Question 3

How to use this scheme. The paper requires Question 1 plus **one** of Question 2 or Question 3 — both are marked out of 50 and count equally toward the paper. This scheme covers both in full so you can revise whichever route you are more comfortable with.

Study-Uni

Question 2

50 Marks — Choice question: answer EITHER Question 2 OR Question 3

(a) Required

8 Marks

Calculate the numbers a)–e) to complete the following table (Present Value, Years, Interest Rate, Future Value):

FULL MODEL ANSWER

Every row uses $FV = PV(1+r)^n$, rearranged for whichever variable is missing.

Present Value	Years	Interest Rate	Future Value
2,500	1	2%	a) ?
4,000	3	8%	b) ?
c) ?	6	3%	80,000
10,000	12	d) ?	30,000
24,000	6	e) ?	41,000

Row	Formula used	Answer
a)	$FV = 2,500 \times (1.02)^1$	2,550.00
b)	$FV = 4,000 \times (1.08)^3$	5,038.85
c)	$PV = 80,000 \div (1.03)^6$	66,998.74
d)	$r = (30,000/10,000)^{1/12} - 1$	9.59%
e)	$r = (41,000/24,000)^{1/6} - 1$	9.34%

Each answer was independently verified in Python using the same $FV = PV(1+r)^n$ relationship rearranged algebraically.

Mark Allocation — Part (a), 8 Marks

Tier	Marks	What separates this tier
7-8	7-8	All five values (a–e) correct, including both rate-solving rows.
5-6	5-6	Three or four values correct.
2-4	2-4	One or two values correct; correct general formula recognised but frequent rearrangement errors.
0-1	0-1	Formula not recognised or applied incorrectly throughout.

Question 2

50 Marks — Choice question: answer EITHER Question 2 OR Question 3

(b) Required

9 Marks

Consider the three stocks outlined below and calculate the stock price for each of the three stocks if the market capitalization rate for each stock is 7.00%.

- i. Stock X is expected to provide a dividend of \$8.00 a share forever. (3 Marks)
- ii. Stock Y is expected to pay a dividend of \$2.50 next year. Thereafter, dividend growth is expected to be 5% a year forever. (3 Marks)
- iii. Stock Z is expected to pay a dividend of \$5.00 next year. Thereafter, dividend growth is expected to be 12% a year for four years (i.e., years 2 through 5 inclusive) and zero thereafter. (3 Marks)

FULL MODEL ANSWER

Stock X — level (no-growth) perpetuity

Price = Dividend ÷ r = \$8.00 ÷ 0.07 = \$114.29

Stock Y — Gordon (constant) growth model

Price = D₁ ÷ (r – g) = \$2.50 ÷ (0.07 – 0.05) = \$125.00

Stock Z — supernormal growth (12% in years 2-5), zero growth thereafter

Year	Dividend
1	\$5.0000
2 (×1.12)	\$5.6000
3 (×1.12)	\$6.2720
4 (×1.12)	\$7.0246
5 (×1.12)	\$7.8676

From year 6 onward the dividend is flat at \$7.8676 forever, so its value at the end of year 5 is a level perpetuity: TV₅ = D₅ ÷ r = 7.8676 ÷ 0.07 = \$112.39

Component	Present Value
PV(D1) through PV(D5), discounted at 7%	\$25.65
PV of terminal value (112.39 ÷ 1.07 ⁵)	\$80.14
Stock Z Price	\$105.79

Each stock uses a valuation model matched to its dividend pattern: level perpetuity (X), Gordon growth (Y), and a finite growth phase folding into a level perpetuity from year 6 (Z) — the same structure as the equivalent question in other papers in this series, with different figures.

Mark Allocation — Part (b), 9 Marks (3 + 3 + 3)

Tier	Marks	What separates this tier
Full marks	3 each	Correct model selected and correctly applied for each stock, with the correct final price.
Partial	1–2 each	Correct model recognised but a computational error (e.g. wrong number of discounting periods for Stock Z's terminal value), own-figure applied thereafter.
Low	1	Dividend correctly identified but wrong valuation model applied.
0	0	No credible attempt for that stock.

Question 2

50 Marks — Choice question: answer EITHER Question 2 OR Question 3

(c) Required

23 Marks

It is argued that the Capital Asset Pricing Model, while useful, has a number of associated problems. Discuss the uses of the CAPM and any associated problems, referencing any empirical research you may be aware of in this area.

FULL MODEL ANSWER

Uses of the CAPM

Estimating the cost of equity: CAPM's most common practical use is to estimate a company's cost of equity capital for use in a weighted average cost of capital (WACC) calculation, which in turn is used as the discount rate in investment appraisal (NPV) decisions.

Benchmarking fund/portfolio performance: CAPM provides a theoretical benchmark expected return given a portfolio's beta, against which realised returns can be compared to assess whether a fund manager has generated genuine outperformance (alpha) or simply taken on more systematic risk.

Pricing risk in a single, intuitive framework: CAPM reduces the pricing of risk to a single factor (beta) and gives a simple, linear relationship between risk and required return, which is easy to communicate and apply consistently across many different investment decisions within an organisation.

Question 2 (continued)

Associated problems

Estimating beta is backward-looking and unstable: Beta is typically estimated from historical regression against a market index, but a company's risk profile changes over time, so a historical beta may poorly estimate the beta that will actually prevail going forward.

Choice of market proxy (Roll's critique): CAPM requires the true market portfolio, but in practice a proxy such as a stock index is used, which excludes bonds, property, and other asset classes. Roll (1977) argued this makes the model effectively untestable.

Empirical evidence of missing risk factors: Fama and French (1992/1993) find that firm size and the book-to-market ratio help explain the cross-section of stock returns beyond market beta alone, motivating multi-factor alternatives to the single-factor CAPM.

Input estimation risk: The risk-free rate and market risk premium both require judgement calls, and the market risk premium in particular varies substantially depending on the historical period and method used to estimate it.

Restrictive assumptions: CAPM assumes frictionless markets, unrestricted risk-free borrowing and lending, no taxes, and homogeneous expectations — assumptions that do not hold precisely in practice.

Candidates may structure this either as two clearly separated sections (uses, then problems) or interwoven, provided both uses AND a well-developed set of problems (with empirical research where possible) are covered. This is a broader question than the equivalent 'limitations only' question in other papers in this series — full marks require the uses section as well as the problems section.

Mark Allocation — Part (c), 23 Marks

Tier	Marks	What separates this tier
18-23	18-23	Clear discussion of at least two to three genuine uses of CAPM AND at least four to five well-explained problems, with reference to empirical research throughout.
12-17	12-17	Uses covered briefly; problems section reasonably developed (three to four points) but empirical research thin.
6-11	6-11	Either uses or problems addressed in isolation (not both), or points listed with little explanation.
0-5	0-5	No accurate discussion of either CAPM's uses or its limitations.

Question 2

50 Marks — Choice question: answer EITHER Question 2 OR Question 3

(d) Required

10 Marks

"There is a significant emphasis on the key role a Finance Manager in an organisation plays" Discuss this statement, describing the role the Finance manager plays in an organisation and the key considerations for a Finance Manager in performing in that role.

FULL MODEL ANSWER

The Finance Manager is responsible for the financial health of the organisation and sits at the centre of the three core corporate finance decisions:

The Investment decision: Identifying and evaluating capital projects and deciding which to undertake, typically using techniques such as NPV and IRR to ensure only value-creating (positive-NPV) projects are pursued.

The Financing decision: Determining how to raise the funds needed to support the investment programme — the mix of debt, equity and retained earnings — while managing the company's overall capital structure and cost of capital.

The Dividend/payout decision: Deciding how much of the company's earnings to distribute to shareholders as dividends (or via buybacks) versus retaining for reinvestment, balancing shareholder income expectations against the company's growth funding needs.

Working capital and liquidity management: Ensuring the organisation has sufficient short-term liquidity to meet its obligations (managing cash, receivables, payables and inventory) without holding excessive idle cash that could otherwise be invested or returned to shareholders.

Key considerations for the Finance Manager

Balancing risk and return: Every decision involves a risk-return trade-off, and the Finance Manager must ensure the level of risk taken on is appropriate given the organisation's objectives and the expectations of shareholders and other stakeholders.

Acting in shareholders' (and stakeholders') interests: Given the agency relationship between managers and shareholders, the Finance Manager must be conscious of aligning decisions with the objective of maximising shareholder wealth while managing relationships with other key stakeholders (employees, creditors, regulators).

Maintaining financial control and reporting integrity: Accurate financial reporting and robust internal controls underpin every other decision the Finance Manager makes, and are essential to maintaining the trust of capital markets, regulators and other stakeholders.

Candidates who structure the answer around the three (or four, including working capital) core financial decisions, plus a discussion of the risk-return and stakeholder considerations that guide the Finance Manager in making them, should receive full credit regardless of specific ordering or emphasis.

Mark Allocation — Part (d), 10 Marks

Tier	Marks	What separates this tier
8-10	8-10	Clear coverage of the investment, financing and dividend decisions (plus ideally working capital), with sound discussion of the key considerations (risk-return, stakeholder alignment).
5-7	5-7	Two of the three core decisions covered adequately; considerations section present but underdeveloped.
2-4	2-4	Only one decision area discussed in any depth, or discussion is generic without reference to the specific role of a Finance Manager.
0-1	0-1	No meaningful discussion of the Finance Manager's role.

Question 2 Total: 50 Marks

Question 3

50 Marks — Choice question: alternative to Question 2

(a) Required

8 Marks

Using the Capital Asset Pricing Model (CAPM) calculate the expected return for Share T, assuming a beta factor of 0.80, a risk free rate of interest of 3%, and a market risk premium of 5%. Explain what is meant by 'market risk' and state if such risk can be diversified away.

FULL MODEL ANSWER

$E(R) = R_f + \beta(\text{Market Risk Premium})$

Component	Value
Risk-free rate, R_f	3.00%
Beta, β	0.80
Market risk premium	5.00%
$\beta \times \text{Market risk premium} = 0.80 \times 5\%$	4.00%
$E(R) = 3\% + 4.00\%$	7.00%

Market risk: Market risk (also called systematic or non-diversifiable risk) is the portion of a share's total risk that arises from economy-wide factors affecting all firms to some degree — interest rates, inflation, GDP growth, geopolitical events — captured in CAPM by beta. It is distinct from firm-specific (unsystematic/idiosyncratic) risk, which arises from factors unique to an individual company.

Can it be diversified away? No — by definition, market risk cannot be eliminated through diversification, because it affects all securities to some degree and so cannot be averaged away by holding many different shares. This is precisely why CAPM only rewards investors for bearing market risk (via beta) and not for bearing firm-specific risk, which a well-diversified investor can eliminate at no cost and therefore is not compensated for.

Mark Allocation — Part (a), 8 Marks

Tier	Marks	What separates this tier
7-8	7-8	Correct CAPM calculation, accurate definition of market risk as systematic/non-diversifiable risk, and a correct, well-explained statement that it cannot be diversified away.
4-6	4-6	Correct calculation but definition of market risk vague, or diversifiability question answered without reasoning.
1-3	1-3	Formula misapplied, or explanation given with no calculation.
0	0	No credible attempt.

Question 3

50 Marks — Choice question: alternative to Question 2

(b) Required9 Marks

A 12-year U.S. Treasury bond with a face value of \$1,000 pays a coupon of 6.00% paid semi-annually. The reported yield to maturity is 3.0%.

i. What is the present value of the bond? (3 Marks)

ii. If the yield to maturity changes to 5%, what would be the present value? (3 Marks)

iii. If the yield to maturity changes to 2%, what would be the present value? (3 Marks)

FULL MODEL ANSWER

Semi-annual coupon = 6.00%/2 × \$1,000 = \$30.00, n = 24 half-years, semi-annual rate = YTM/2.

Price = C × [1 − (1+r)^{−n}] / r + FV / (1+r)ⁿ

Yield to Maturity	PV of Coupons	PV of Face Value (\$1,000)	Bond Price
3.0% (given)	600.91	699.54	\$1,300.46
5.0%	536.55	552.88	\$1,089.42
2.0%	637.30	787.57	\$1,424.87

The coupon rate (6.00%) exceeds every YTM tested, so the bond trades at a premium to face value throughout — consistent across all three prices. The inverse YTM/price relationship holds: price falls as YTM rises from 2% → 3% → 5%.

Mark Allocation — Part (b), 9 Marks (3 per YTM scenario)

Tier	Marks	What separates this tier
3 (per scenario)	3	Correct bond price with semi-annual coupon and rate correctly applied.
1-2 (per scenario)	1-2	Correct formula structure but an arithmetic error, or annual compounding used instead of semi-annual (own-figure).
0 (per scenario)	0	Wrong formula or no attempt.

Question 3

50 Marks — Choice question: alternative to Question 2

(c)(i) Required8 Marks

An investor believes that shares in Candy Plc are currently underpriced at €30.00 and that they will trade significantly higher in three months' time. The premium on a three-month Candy Plc Call option with an exercise price of €33.00 is currently €1.50.

1) Given the information provided, indicate the option strategy the investor would employ to take advantage of any share price increase.

2) Calculate the profit/(loss) that would accrue from such a strategy if Candy Plc's share price after three months was: (a) €27.50 (b) €37.50. Supplement your answer with appropriate payoff diagrams.

FULL MODEL ANSWER

Strategy: The investor should buy a call option on Candy Plc with a €33.00 exercise price (a **long call**). This gives the right, but not the obligation, to buy the shares at €33.00 — if the investor's view is correct and the price rises well above €33.00, the call captures that upside for a much smaller upfront outlay (the €1.50 premium) than buying the shares outright.

Profit/loss on the call option, at expiration, per share:

Profit = Max(Share Price – Exercise Price, 0) – Premium Paid

Share Price at Expiry	Call exercised?	Intrinsic Value	Less: Premium Paid	Profit/(Loss) on Call
€27.50	No — price below strike	€0.00	€1.50	(€1.50)
€37.50	Yes — buy at €33 vs market €37.50	€4.50	€1.50	€3.00

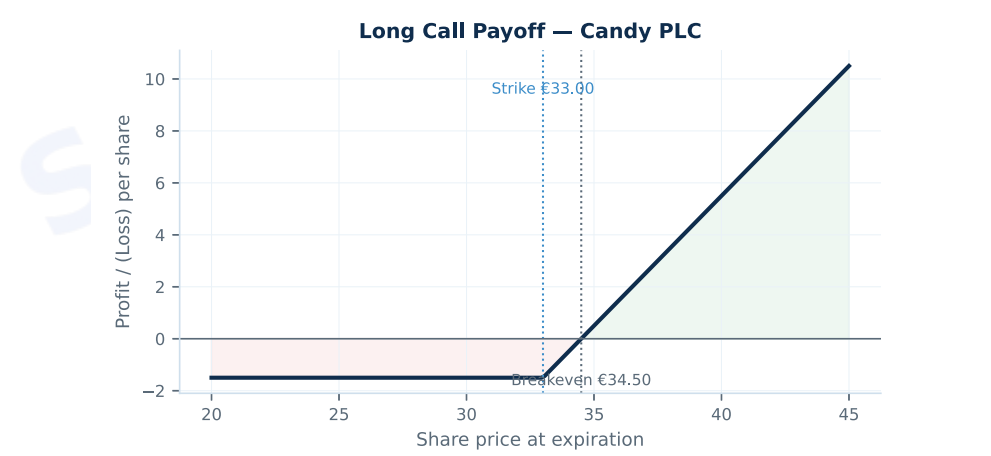


Fig. 1 — Long call payoff, Candy PLC (strike €33.00, premium €1.50, breakeven €34.50)

At €27.50 the option expires worthless (share price is below the €33.00 strike), so the investor's loss is capped at the €1.50 premium paid — this capped-downside, uncapped-upside profile is the defining characteristic of a long call. At €37.50 the option is comfortably in the money and the investor profits €3.00 per share, far more than the return available from simply holding the shares given the small initial outlay. The breakeven share price is $\text{€}33.00 + \text{€}1.50 = \text{€}34.50$.

Mark Allocation — Part (c)(i), 8 Marks

Tier	Marks	What separates this tier
7-8	7-8	Correct strategy identified (buy a call / long call) with clear reasoning; both profit/loss figures correct; payoff diagram correctly shaped (kinked at strike, breakeven at €34.50).
5-6	5-6	Correct strategy and both figures correct, but diagram missing or materially incorrect.
2-4	2-4	Correct strategy identified but one or both profit/loss figures wrong, or a put option used instead of a call.
0-1	0-1	Wrong strategy or no credible attempt.

Question 3

50 Marks — Choice question: alternative to Question 2

(c)(ii) Required

15 Marks

There are a number of factors that affect option prices as specified in the Black-Scholes option pricing formula. Detail these factors and briefly outline the affect of a change in each of these factors on the prices of both Put and Call options.

FULL MODEL ANSWER

Variable	Effect on Call price	Effect on Put price
Current share price (S)	Increases as S rises	Decreases as S rises
Exercise/strike price (K)	Decreases as K rises	Increases as K rises
Time to expiration (T)	Increases with more time	Generally increases with more time
Volatility of the underlying (σ)	Increases with volatility	Increases with volatility
Risk-free interest rate (r)	Increases as r rises	Decreases as r rises

Share price and strike price: A call becomes more valuable the further the share price is above the strike (more intrinsic value); a put becomes more valuable the further the share price is below the strike — the direct in-the-money effect for each option type. This is the mechanism illustrated by the Candy PLC call above: its value rises as the share price moves further above the €33.00 strike.

Time to expiration: More time gives more opportunity for the underlying price to move favourably before expiry, so both calls and puts on non-dividend-paying shares are generally worth more with more time remaining (time value); this time value decays toward zero as expiration approaches (time decay).

Volatility: Because option payoffs are asymmetric — losses capped at the premium paid, upside uncapped for the holder — higher volatility raises the probability of a large favourable move without symmetrically raising the cost of a large unfavourable one, so both call and put values rise with volatility. This is why an investor's belief that a share is likely to move significantly (as with Candy PLC) makes buying an option an attractive way to express that view.

Risk-free interest rate: A higher rate raises the cost of carrying the underlying and lowers the present value of the exercise price paid in the future, which increases call values (the strike is effectively 'cheaper' in present-value terms) and decreases put values.

Dividends (where applicable): Although not one of the five core Black-Scholes variables in the basic (non-dividend-paying) model, expected dividends on the underlying share reduce call values and increase put values, since the share price is expected to fall by roughly the dividend amount on the ex-dividend date.

Candidates who correctly identify the five core variables and their qualitative direction of effect on both call and put prices should receive full credit. Discussion of dividends as an extension is a bonus point rather than a requirement, since the standard Black-Scholes formula assumes no dividends.

Mark Allocation — Part (c)(ii), 15 Marks

Tier	Marks	What separates this tier
12-15	12-15	All five variables correctly identified with accurate direction of effect on BOTH calls and puts, sound underlying reasoning, and clear linkage to option pricing intuition.
8-11	8-11	Four or five variables identified with mostly correct direction, minor errors on one variable.
4-7	4-7	Two to three variables correctly identified, or direction of effect frequently confused between calls and puts.
0-3	0-3	Fewer than two variables correctly identified.

Question 3

50 Marks — Choice question: alternative to Question 2

(d) Required

10 Marks

Discuss the forms that the Financing Decision can take for an organisation and the criticality of the Financing Decision to the long term success of the organisation.

FULL MODEL ANSWER

The financing decision concerns how a company raises the funds needed to support its investment decisions:

Equity finance: Raising funds by issuing ordinary shares, giving shareholders a residual claim and voting rights but no obligation to pay a fixed return — the most flexible but typically most expensive source of capital.

Debt finance: Borrowing via bank loans, bonds, or other fixed-income instruments, carrying a contractual obligation to pay interest and repay principal, generally cheaper than equity due to lower risk to the provider and tax-deductibility of interest.

Retained earnings: Using internally generated cash flow, avoiding issuance costs and dilution, but limited by the scale of the company's own profitability.

Hybrid instruments: Convertible bonds or preference shares combine features of debt and equity, allowing companies to tailor the risk/return and control characteristics of the capital raised.

Why the financing decision is critical to long-term success

Capital structure drives the cost of capital: The mix of debt and equity chosen directly determines the firm's weighted average cost of capital (WACC), which is the discount rate used to evaluate every future investment — getting this wrong systematically biases investment decisions.

Financial risk and solvency: Excessive reliance on debt raises fixed obligations that must be met regardless of trading conditions, increasing the risk of financial distress or insolvency in a downturn; too little debt, conversely, may mean the firm is not using cheaper capital efficiently and is foregoing the tax shield on interest.

Financing constrains strategic flexibility: A poorly structured financing decision (e.g. over-reliance on short-term debt for long-term assets, breaching the matching principle) can leave a company unable to fund growth opportunities or refinance at reasonable cost when market conditions tighten, directly threatening long-term competitiveness.

Signalling effects: Financing choices are interpreted by the market as signals about management's confidence in the company's prospects (e.g. new equity issuance is often read as a signal that management believes shares are overvalued), so the financing decision also has reputational and share-price consequences beyond its direct cost.

Candidates may instead or additionally discuss the maturity-matching principle (financing long-term assets with long-term liabilities) or specific examples of financial distress arising from poor financing choices. Any well-explained combination of financing forms plus a reasoned discussion of criticality should receive full credit.

Mark Allocation — Part (d), 10 Marks

Tier	Marks	What separates this tier
8-10	8-10	Clear coverage of equity, debt and retained earnings (or hybrids) as financing forms, PLUS a well-developed discussion of why the decision is critical to long-term success (cost of capital, financial risk, strategic flexibility).
5-7	5-7	Financing forms covered adequately but the 'criticality' discussion underdeveloped or missing.
2-4	2-4	Only one financing form discussed, or discussion generic without addressing criticality.
0-1	0-1	No meaningful discussion.

Question 3 Total: 50 Marks